■ Developer Mock Test Results

Student: **Shiva kasula**

Name:	Shiva kasula	**Student ID:**	112
Date:	2026-03-11 12:41:37	**Performance:**	■■ Needs Improvement
Overall Score:	5.0/25 (20.0%)	**Test ID:**	78c84641-c77f-4664...
Warnings:	0/3	**Status:**	Completed ■

■ Section-wise Performance

Section	Score	Percentage	Status
■ Aptitude	3/10	30.0%	■■ Needs Work
■ MCQ/Theory	2/10	20.0%	■■ Needs Work
■ Coding	0.0/5	0.0%	■■ Needs Work

■ Detailed Question Review

■ APTITUDE SECTION (3/10 - 30.0%)

■ Q1. A company has 100 employees, and the average salary is ■50000. If 10 new employees are hired, what is the new average salary? The total salary of the...

Your Answer: ■47500

■ *Excellent! Right answer.*

■ Q2. A sum of money is invested at a compound interest rate of 10% per annum for 3 years. If the interest is compounded annually, what is the amount of mo...

Your Answer: ■13610

Correct Answer: ■13310

■ *The correct answer is ■13310 because the principal amount of ■10000 is invested at a compound interest rate of 10% per annum for 3 years. The key calculation step is: Amount = 10000 * (1 + 0.10)^3 = 10000 * (1.10)^3 = 10000 * 1.331 = ■13310. The user's likely mistake is a calculation error, as their answer ■13610 does not match the correct calculation.*

■ Q3. A boat travels 24 kilometers upstream in 4 hours and 36 kilometers downstream in 3 hours. What is the speed of the boat in still water? The speed of ...

Your Answer: 8 km/h

■ *Correct! Good understanding.*

■ Q4. Find the next number in the series: 3, 6, 12, 24, 48, ? The pattern of the series is obtained by multiplying the previous term by 2. Select the correc...

Your Answer: 64

Correct Answer: 192

■ *The correct answer is actually 96, not 192. The pattern of the series is obtained by multiplying the previous term by 2, so the key calculation step is: 48 * 2 = 96. The user's likely mistake was not applying the multiplication pattern correctly to find the next number in the series.*

■ Q5. A and B started a business with investments of ■50000 and ■75000 respectively. If the profit is ■30000, what is the share of A? The ratio of their in...

Your Answer: ■10000

Correct Answer: ■12500

■ *The correct answer is ■12500 because the ratio of their investments is 5:7.5, which can be simplified to 2:3, and the total parts is 2+3 = 5. To find A's share, we calculate (2/5) * ■30000 = ■12000, but since the ratio of investments is given as 5:7.5, we use this ratio to find A's share: (5/(5+7.5)) * ■30000 = (5/12.5) * ■30000 = ■12000. However, considering the given options and the standard method of calculating the share based on the investment ratio, the calculation should be: (5/12.5) * ■30000 = (2/5) * ■30000 = ■12000, but using the exact investment amounts: A's share = (50000 / (50000 + 75000)) * ■30000*

■ Q6. Pointing to a photograph, Rohan says, "She is the daughter of my grandfather's only son." Who is the girl in the photograph? Rohan's grandfather has ...

Your Answer: Rohan's mother

■ *Excellent! Right answer.*

■ Q7. In how many ways can 5 people be arranged in a row? The order of arrangement matters. Select the correct answer from the options below.

Your Answer: 30

Correct Answer: 120

■ *The correct answer is 120 because the number of ways to arrange 5 people in a row is given by the factorial of 5, denoted as 5!. The key calculation step is: 5! = 5 × 4 × 3 × 2 × 1 = 120. The user's answer of 30 is likely incorrect because they may have only considered a limited number of arrangements or made a calculation error, whereas the factorial calculation provides the total number of possible arrangements.*

■ Q8. If 'house' is coded as 'esuoh', what is the code for 'tree'? Each letter is shifted by a fixed number of positions. Select the correct answer from the...

Your Answer: No answer (Skipped)

Correct Answer: eetr

■ *The correct answer is 'eetr' because the coding pattern involves reversing the order of the letters in the word. In the given example, 'house' is coded as 'esuoh', which is the reverse of 'house'. Therefore, applying the same pattern to 'tree', we get 'eert', which is the correct code.*

■ Q9. If sin(x) = 3/5, what is the value of cos(x)? The trigonometric identity sin^2(x) + cos^2(x) = 1 can be used. Select the correct answer from the optio...

Your Answer: No answer (Skipped)

Correct Answer: 4/5

■ *To find the value of cos(x), we can use the identity sin^2(x) + cos^2(x) = 1. Given sin(x) = 3/5, we can substitute this value into the equation: (3/5)^2 + cos^2(x) = 1. The key calculation step is: (3/5)^2 = 9/25, so 9/25 + cos^2(x) = 1, which simplifies to cos^2(x) = 1 - 9/25 = 16/25, and taking the square root of both sides gives cos(x) = 4/5. The correct answer is indeed 4/5. If the user got it wrong, their likely mistake was incorrectly applying the trigonometric identity or making a calculation error when solving for cos(x).*

■ Q10. Find the next number in the series: 2, 6, 12, 20, 30, ? The pattern of the series is obtained by adding 4, 6, 8, 10, ... Select the correct answer fro...

Your Answer: No answer (Skipped)

Correct Answer: 42

■ *The correct answer is 42. The pattern of the series is obtained by adding consecutive even numbers: +4 (2 to 6), +6 (6 to 12), +8 (12 to 20), +10 (20 to 30), and +12 (30 to 42). The key calculation step is 30 + 12 = 42, which confirms that the next number in the series is indeed 42.*

■ MCQ SECTION (2/10 - 20.0%)

■ Q1. What file extension is used for Go programs? Choose the correct answer from the options below.

Your Answer: .go

■ *Well done!*

■ Q2. What is the name of the company that developed the Go programming language, according to the guide? Choose the correct answer from the options below.

Your Answer: Microsoft

Correct Answer: Google

■ *The correct answer is Google because the Go programming language, also known as Golang, was indeed developed by Google. The user's answer, Microsoft, is incorrect, and their likely mistake was confusing Google with another major tech company. Google's involvement in the development of the Go language is well-documented, and it was designed by Robert Griesemer, Rob Pike, and Ken Thompson at Google.*

■ Q3. What is the type of software that Go is designed for building, according to the guide? Choose the correct answer from the options below.

Your Answer: No answer (Skipped)

Correct Answer: Fast and reliable software

■ *The correct answer is A) Fast and reliable software. This is because the Go programming language is designed to be efficient, simple, and easy to use, making it ideal for building software that requires speed and reliability. The user likely chose the correct answer because the guide emphasizes Go's ability to compile quickly and run efficiently, which are key characteristics of fast and reliable software.*

■ Q4. What are the fundamental concepts used to build Go programs, as listed in the guide? Choose the correct answer from the options below.

Your Answer:

```
No answer (Skipped)
```

Correct Answer:

```
Variables, data types, control structures, functions, structs, and packages
```

■ *The correct answer is B) Variables, data types, control structures, functions, structs, and packages. This is because Go programs are built using these fundamental concepts, with structs serving as the primary means of organizing data and packages enabling code organization and reuse. The user's likely mistake was not including structs and packages, which are unique to Go and essential for building robust programs. (Note: The user's answer was not provided, but the explanation assumes they chose an incorrect option.)*

■ Q5. What is a characteristic of functions in Go, according to the guide? Choose the correct answer from the options below.

Your Answer: No answer (Skipped)

Correct Answer: They are blocks of reusable code that perform specific tasks

■ *The correct answer is A) They are blocks of reusable code that perform specific tasks. This is because functions in Go are designed to be reusable, allowing developers to organize and structure their code in a modular way, making it easier to maintain and update. The user likely chose the correct answer because it accurately reflects the fundamental purpose of functions in programming, which is to encapsulate specific tasks that can be called multiple times from different parts of the code.*

■ Q6. What is the term used to describe a true or false value in Go programming? Choose the correct answer from the options below.

Your Answer: Float

Correct Answer: Boolean

■ *The correct answer is Boolean because in Go programming, a boolean is a data type that can have one of two values: true or false. The user's answer, Float, is incorrect because a float is a data type used to represent decimal numbers, not true or false values. The user likely mistook the term because they may not be familiar with the specific data types in Go programming, leading them to choose a type that is used for numerical values instead of logical values.*

■ **Q7. What is the term used to describe a function that has access to its own scope and the scope of its outer functions in Go programming? Choose the corre...**

Your Answer: No answer (Skipped)
Correct Answer: Closure
■ *The correct answer is A) Closure. This is because a closure in Go programming refers to a function that has access to its own scope and the scope of its outer functions, allowing it to "remember" and use variables from those outer scopes even when the outer functions have returned. The user likely chose correctly, but if they had chosen incorrectly, their mistake would likely be due to confusing the term "closure" with related but distinct concepts like anonymous functions or higher-order functions.*

■ **Q8. What is a characteristic of the Go standard library, according to the guide? Choose the correct answer from the options below.**

Your Answer:

```
It is only available for certain platforms
```

Correct Answer:

```
It is large and comprehensive
```

■ *The correct answer, "It is large and comprehensive", is right because the Go standard library is indeed known for its extensive range of packages and functionalities that support various aspects of programming, making it a robust and versatile tool for developers. The user's answer, "It is only available for certain platforms", is incorrect because the Go standard library is designed to be platform-independent, allowing it to be used across different operating systems and environments. The user likely mistook the Go standard library's capabilities, underestimating its breadth and cross-platform compatibility.*

■ **Q9. What is the term used to describe a dynamic array in Go programming? Choose the correct answer from the options below.**

Your Answer: Slice
■ *Correct! Good understanding.*

■ **Q10. What is the term used to describe the ability to assign functions to variables, pass them as arguments, and return them from other functions in Go pro...**

Your Answer:

```
Functions as second-class citizens
```

Correct Answer:

```
Functions as first-class citizens
```

■ *The correct answer is "Functions as first-class citizens" because in Go programming, functions can be assigned to variables, passed as arguments to other functions, and returned from functions, which are the characteristics of first-class citizens. This means that functions are treated as objects and can be manipulated like any other value. The user's answer, "Functions as second-class citizens", is likely incorrect because it implies that functions have limited rights or capabilities, which is not the case in Go programming where functions have the full range of capabilities.*

■ CODING SECTION (0.0/5 - 0.0%)

■ **Q1. Write a Go program that reads a string from stdin and attempts to parse it as a hexadecimal color code. If the parsing fails, print an error message. ...**

Your Answer:

```
package main

import "fmt"

// Write your solution here
func solution() {
    // Your code here
    fmt.Println("Hello, World!")
```

```
}

func main() {
    solution()
}
```

Correct Answer:

```
package main

import (
"bufio"
"fmt"
"os"
"regexp"
)

func main() {
reader := bufio.NewReader(os.Stdin)
input, _ := reader.ReadString('\n')
input = removeNewLine(input)

regex := regexp.MustCompile(`^#([A-Fa-f0-9]{6}|[A-Fa-f0-9]{3})$`)
if regex.MatchString(input) {
fmt.Println(input)
} else {
fmt.Println("Invalid hexadecimal color code")
}
}

func removeNewLine(str string) string {
return string(str[:len(str)-1])
}
```

■ It looks like your code isn't quite reading the input from stdin as required by the problem, so let's start by looking into how to use the `fmt.Scanln` function to read a string from the user. To get started, you can try using `fmt.Scanln` to read the input string and store it in a variable, then use the `fmt.Sprintf` function to attempt to parse it as a hexadecimal color code, and Keep practicing, you will get there! ■ Score: 0.0 / 1

■ Q2. Write a Go program that reads a string from stdin and checks if it's a valid email address. **Input Format:** - Line 1: A string **Output Format:** ...

Your Answer:

```
package main

import "fmt"

// Write your solution here
func solution() {
    // Your code here
    fmt.Println("Hello, World!")
}

func main() {
    solution()
}
```

Correct Answer:

```
package main

import (
"bufio"
"fmt"
```

```
"os"
"regexp"
)

func main() {
reader := bufio.NewReader(os.Stdin)
email, _ := reader.ReadString('\n')
email = email[:len(email)-1] // remove newline character

matched, _ := regexp.MatchString(`^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$`, email)
if matched {
fmt.Println("Valid")
} else {
fmt.Println("Invalid")
}
}
```

■ *It looks like your code is currently printing a greeting message instead of checking the email address, so let's start by reading the input from stdin and storing it in a variable to check if it's a valid email address. You can begin by using the `fmt.Scanln` function to read the input string from the user, and then use a regular expression or a simple string manipulation to validate the email address format, Keep practicing, you will get there! ■ Score: 0.0 / 1*

■ Q3. Write a Go program that reads an array of integers from stdin and prints the sum of all values. **Input Format:** - Line 1: An integer n, the size of...

Your Answer:

```
package main

import "fmt"

// Write your solution here
func solution() {
    // Your code here
    fmt.Println("Hello, World!")
}

func main() {
    solution()
}
```

Correct Answer:

```
package main

import "fmt"

func main() {
    var n int
    fmt.Scan(&n)
    sum := 0
    for i := 0; i < n; i++ {
        var num int
        fmt.Scan(&num)
        sum += num
    }
    fmt.Println(sum)
}
```

■ *It looks like your code is currently printing a greeting message instead of the sum of the array values, so let's start by reading the input from stdin and storing the array values in a variable. To get started, you can use the `fmt.Scanln` function to read the array size and values from the input, and then calculate the sum of these values. Keep practicing, you will get there! ■ Score: 0.0 / 1*

■ **Q4. Write a Go program that reads a string from stdin and checks if it's a valid phone number (in the format XXX-XXX-XXXX). **Input Format:** - Line 1: A...**

Your Answer:

```
package main

import "fmt"

// Write your solution here
func solution() {
    // Your code here
    fmt.Println("Hello, World!")
}

func main() {
    solution()
}
```

Correct Answer:

```
package main

import (
"bufio"
"fmt"
"os"
"regexp"
)

func main() {
reader := bufio.NewReader(os.Stdin)
line, _ := reader.ReadString('\n')
line = line[:len(line)-1] // remove newline character

matched, _ := regexp.MatchString(`^\d{3}-\d{3}-\d{4}$`, line)
if matched {
fmt.Println("Valid")
} else {
fmt.Println("Invalid")
}
}
```

■ *It looks like your code is currently printing a greeting message instead of checking the phone number, so let's start by reading the input string from stdin and storing it in a variable to analyze its format. To get started, you can use the `fmt.Scanln()` function to read the input string and then use conditional statements to check if it matches the valid phone number pattern, Keep practicing, you will get there!* ■ *Score: 0.0 / 1*

■ **Q5. Write a Go program that reads a string from stdin representing a shopping cart item (name and price) and calculates the total cost. **Input Format:**...**

Your Answer:

```
package main

import "fmt"

// Write your solution here
func solution() {
    // Your code here
    fmt.Println("Hello, World!")
}

func main() {
    solution()
}
```

Correct Answer:

```
package main

import (
"bufio"
"fmt"
"os"
"strconv"
"strings"
)

func main() {
reader := bufio.NewReader(os.Stdin)
var n int
fmt.Scanln(&n)

totalCost := 0.0
for i := 0; i < n; i++ {
var item string
fmt.Scanln(&item)
parts := strings.Split(item, ":")
if len(parts) != 2 {
fmt.Println("Invalid input format")
return
}
price, err := strconv.ParseFloat(strings.TrimSpace(parts[1]), 64)
if err != nil {
fmt.Println("Invalid price")
return
}
totalCost += price
}
fmt.Println(totalCost)
}
```

■ *It looks like your code is currently printing a greeting message instead of calculating the total cost of the shopping cart items, so let's start by reading the input from stdin and parsing the item names and prices. To get started, you can use the `bufio` package to read lines from stdin and then split each line into the item name and price using the `strings.Split` function, and then Keep practicing, you will get there!* ■ *Score: 0.0 / 1*

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.
- **Coding:** Practice more coding problems on platforms like LeetCode or HackerRank.
